Rigid motion: geometry first

Use the laboratory ground frame unless stated. In planar motion, +z points out of the sheet.

One body, one angular speed

Rigidity means all pairwise distances stay fixed. Translation carries the centre of mass C; rotation describes motion relative to C.

$$\vec{v}_P = \vec{v}_C + \vec{\omega} \times \vec{r}_{P/C}$$

WHY / The relative velocity is perpendicular to the line joining two body points: no change in their separation.

WATCH / Equal angular speed does not mean equal linear speed. Distance is measured perpendicular to the rotation axis.

$$\vec{v}_B - \vec{v}_A = \vec{\omega} \times \vec{r}_{B/A}$$

For any two points A and B on the body, their relative velocity has no component along AB.

Fixed-axis kinematics

$$\omega = \dot{\theta}, \quad \alpha = \dot{\omega}$$

$$\omega = \omega_0 + \alpha t$$

$$\Delta\theta = \omega_0 t + \tfrac{1}{2}\alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$$

WHY / The last three equations require constant alpha about a fixed axis. Use signed angles in radians.

SI: omega in rad/s; alpha in rad/s². Convert rpm using omega = 2 pi n / 60.

$$\alpha = \omega\, d\omega/d\theta$$

For variable alpha, integrate d(omega)=alpha dt or omega d(omega)=alpha d(theta); do not use constant-alpha shortcuts.

Acceleration is not tangent only

$$\vec{a}_P = \vec{a}_C + \vec{\alpha} \times \vec{r}$$
$$+ \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

Here r = r(P/C). For a fixed axis: tangential acceleration = alpha r; inward normal acceleration = omega² r.

WHY / Even constant angular velocity changes the direction of each off-axis point's velocity.

WATCH / Do not replace total acceleration by alpha r; the inward term remains.

$$a = \sqrt{(\alpha r)^2 + (\omega^2 r)^2}$$

The two components are perpendicular for fixed-axis circular motion. At the axis both are zero.

Instantaneous centre

For nonzero omega in planar motion, locate the instantaneous centre by intersecting lines normal to two known velocities.

$$v_P = |\omega|\, CP_{\text{inst}}$$

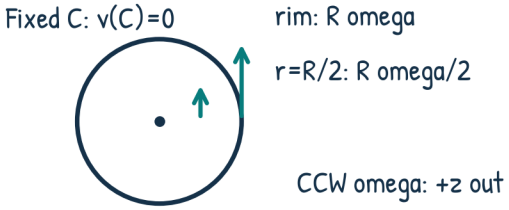
WHY / It is a velocity construction, not a permanent hinge. Pure translation has its centre at infinity.

WATCH / Do not automatically use fixed-pivot acceleration or torque equations about this moving centre.

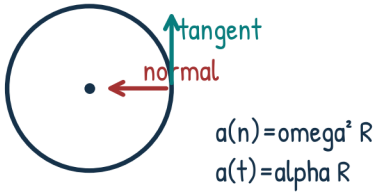
$$\omega = \frac{v_B - v_A}{AB}$$

This signed shortcut needs parallel velocities perpendicular to AB and a consistent direction. Otherwise use the vector relation.

Fixed-axis rotation: tangent velocity and inward acceleration



Fixed-axis rotation: tangent velocity and inward acceleration



REMEMBER / A point can have zero instantaneous velocity without having zero acceleration.

MOI I: rods, plates & axes

Learn the standard results with their axes; the integral explains each coefficient. Uniform bodies unless stated.

Definition: weighted distance²

$$I_A = \int r_\perp^2 \, dm, \quad I_A = Mk_A^2$$

$$K = \tfrac{1}{2} I_A \omega^2$$

k is the radius of gyration about A. For fixed-axis rotation each mass has speed omega r(perp), so summing dm v²/2 proves the energy result.

WHY / Distant mass matters quadratically, not linearly. I depends on shape and axis, not angular speed.

$$0 \le I_A \le MR_{\max}^2$$

The bound assumes every element is within distance R(max) from A. Units: kg m².

Thin rod: centre, end, offset

Length l, mass M; x along the rod measured from its midpoint. All axes here are perpendicular to the rod.

$$dm = (M/l)dx$$

$$I_C = \frac{M}{l}\int_{-l/2}^{l/2} x^2 dx = \frac{Ml^2}{12}$$

$$I_A = M(l^2/12 + d^2)$$

$$I_{\text{end}} = Ml^2/3$$

WHY / Shift from the centre by d. An end axis has d=l/2, so 1/12+1/4=1/3.

WATCH / A zero-thickness rod about its own length has I=0; a real thick rod does not.

Rectangular and square plates

Rectangle a along x, b along y; origin C. x,y lie in the plate, z is normal.

$$I_x = Mb^2/12, \quad I_y = Ma^2/12$$

$$I_z = M(a^2 + b^2)/12$$

Derive Ix by strips parallel to x: dm=(M/b)dy; integrate y² dm. Use the perpendicular-axis theorem for Iz.

$$I_{\text{edge}\,\parallel\,\text{x}} = Mb^2/3$$

$$a = b = s: \; I_z = Ms^2/6$$

WHY / For a square any central in-plane axis, including a diagonal, has I=Ms²/12. Symmetry removes the cross term.

Inclined axis through the rod

Axis through C makes angle phi with a thin rod. A point at signed coordinate x has perpendicular distance |x sin(phi)|.

$$I_C(\phi) = \frac{Ml^2}{12}\sin^2\phi$$

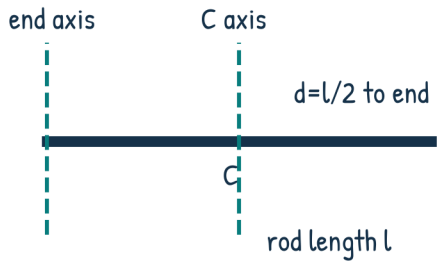
$$I_{\text{end}}(\phi) = \frac{Ml^2}{3}\sin^2\phi$$

WHY / Insert r(perp)=|x sin(phi)| into the integral, not x itself. Parallel-axis shifts must stay perpendicular to the chosen axis.

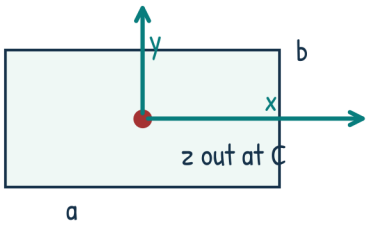
Checks: phi=0 gives zero for a line rod; phi=90° gives the usual transverse result.

WATCH / Do not use a simple sin² rule for a finite-width plate without accounting for its other dimension.

Rod: both marked axes are perpendicular to its length



Rectangle axes: a along x; b along y; z normal



REMEMBER / A moment of inertia without an axis is an incomplete result. Read each formula together with its geometry.

MOI II: rings, discs & annuli

Here z is the central normal to the lamina. A diameter is an in-plane axis through the circle centre O.

Thin ring: learn three axes

$$I_z = MR^2$$

Every element is the same distance R from the normal axis, so the integral immediately equals R² times M.

$$I_{\text{diameter}} = MR^2/2$$

WHY / Two perpendicular diameters are equivalent by symmetry and their inertias add to Iz.

$$I_{\text{tangent, in-plane}} = 3MR^2/2$$

$$I_{\text{rim normal}} = 2MR^2$$

WATCH / "Tangent" alone does not specify whether the axis is in the plane or normal to it.

Disc: concentric-ring derivation

$$\sigma = M/(\pi R^2), \quad dm = 2\pi\sigma r\,dr$$

$$I_z = 2\pi\sigma\int_0^R r^3 dr = MR^2/2$$

$$I_{\text{diameter}} = MR^2/4$$

$$I_{\text{rim normal}} = 3MR^2/2$$

$$I_{\text{tangent, in-plane}} = 5MR^2/4$$

WHY / A disc includes rings of every radius: it has less axial inertia than a ring of the same M and R.

Radius of gyration: k(z)=R/sqrt(2), k(diameter)=R/2.

Annulus: subtract or integrate

Inner radius a, outer radius b, mass M. Uniform areal density sigma=M/[pi(b²-a²)].

$$I_z = 2\pi\sigma\int_a^b r^3 dr$$

$$I_z = \tfrac{1}{2}M(a^2+b^2)$$

$$I_{\text{diameter}} = \tfrac{1}{4}M(a^2+b^2)$$

WHY / Use b⁴-a⁴=(b²-a²)(b²+a²). Alternatively subtract two discs with their original common density.

Checks: a=0 gives a disc. As a->b at fixed mass the annulus approaches a thin ring.

WATCH / M is the remaining annulus mass, not the mass of the uncut outer disc.

Half-circle: O is not its COM

For a semicircular disc: O is the original circle centre; C lies on its symmetry line above the diameter.

$$OC = 4R/(3\pi), \quad I_{O,z} = MR^2/2$$

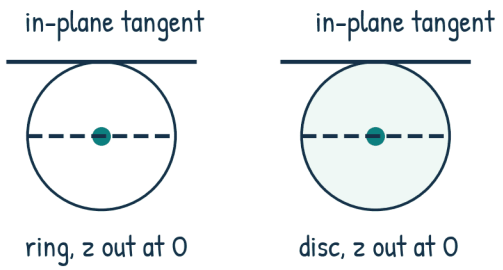
$$I_{C,z} = MR^2/2 - M(4R/(3\pi))^2$$

For a semicircular wire instead: OC=2R/pi and I(O,z)=MR²; subtract M(OC)² to get I(C,z).

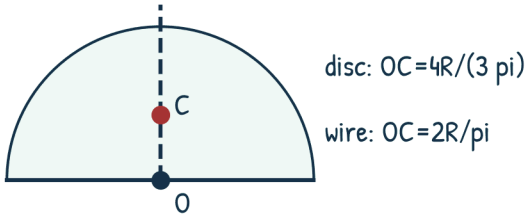
WHY / Cutting a uniform circle in half leaves the average r² about O unchanged; the new COM is displaced.

WATCH / These are derivable applications, not a reason to memorise every cut shape. Always distinguish O from C.

Diameter dashed; tangent solid; normal axis shown by a dot



Half-disc and half-wire: their COMs are not at O



REMEMBER / Normal-axis and diameter-axis results differ by a factor of two for a complete circular lamina or thin ring.

MOI III: cylinders & spheres

Axial = cylinder symmetry axis; transverse = perpendicular through C; a sphere diameter passes through C.

Solid cylinder: both directions

Uniform cylinder radius R, length l. For its symmetry axis each thin disc contributes half dm R².

$$I_{\text{axial}} = MR^2/2$$

$$I_{\text{transverse}} = M(3R^2 + l^2)/12$$

$$dI_{\text{transverse}} = (R^2/4 + z^2)dm$$

WHY / Each slice contributes its own diameter inertia plus its parallel-axis shift. Integrate z from -l/2 to l/2 with dm=(M/l)dz.

l->0 gives a disc diameter; R->0 gives a thin rod transverse to its length.

Hollow cylinder: inner radius a

Outer radius b, length l, total remaining mass M. Uniform material, common central axis.

$$I_{\text{axial}} = M(a^2 + b^2)/2$$

$$I_{\text{transverse}} = M[3(a^2 + b^2) + l^2]/12$$

WHY / Derive by subtracting solid coaxial cylinders at the same density, or replace each slice by an annular lamina.

$$\text{thin wall}: \; I_{\text{axial}} = MR^2$$

$$I_{\text{transverse}} = M(6R^2 + l^2)/12$$

WATCH / A thin cylindrical wall has no end caps in this model. Its transverse inertia is not that of a thin ring unless l->0.

Solid sphere: derive by discs

Let x run along the diameter axis; slice radius s satisfies s²=R²-x². Density rho=3M/(4 pi R³).

$$dm = \rho\pi(R^2 - x^2)dx$$

$$I = \frac{\rho\pi}{2}\int_{-R}^{R}(R^2 - x^2)^2 dx$$

$$I_C = 2MR^2/5, \quad I_{\text{tangent}} = 7MR^2/5$$

WHY / A disc slice rotates about its own normal: dI=dm s²/2, not dm s².

Spherical symmetry makes every central diameter equivalent. k(C)=sqrt(2/5) R.

Spherical shells: thin and thick

Thin uniform shell: a latitude strip at polar angle theta has dm=(M/2)sin(theta)d(theta) and r(perp)=R sin(theta).

$$I = \frac{MR^2}{2}\int_0^{\pi}\sin^3\theta\, d\theta$$

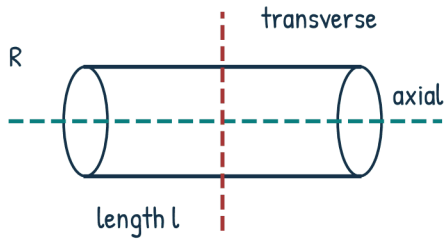
$$I_{\text{thin}} = 2MR^2/3$$

$$I_{\text{thick}} = \frac{2M}{5}\frac{b^5 - a^5}{b^3 - a^3}$$

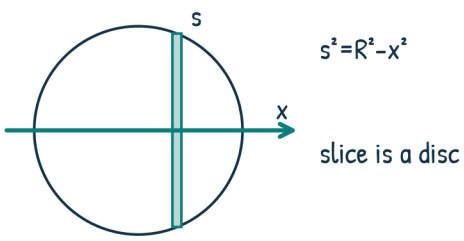
WHY / The thick-shell formula subtracts two same-density solid spheres, inner a and outer b. a->0 gives solid; a->b gives thin shell.

WATCH / A sphere has no special normal axis like a disc; specify a diameter or a displaced parallel line.

Cylinder: distinguish central longitudinal and transverse axes



Sphere sliced normal to a diameter: dI = dm s² / 2



REMEMBER / Do not apply the perpendicular-axis theorem to a 3-D body. Slice it into thin laminae and integrate instead.

MOI IV: composites & integrations

Memorise the standard families; derive unfamiliar density profiles and cut shapes from a few reliable moves.

An off-centre circular hole

Full disc mass M0, radius R; hole radius a, centre offset d. Hole fits inside the disc. Both densities are unchanged.

$$m_h = M_0 a^2/R^2, \quad M = M_0 - m_h$$

$$I_{O,z} = M_0 R^2/2 - m_h(a^2/2 + d^2)$$

$$x_C = -m_h d/M$$

$$I_{C,z} = I_{O,z} - M x_C^2$$

WHY / Take +x toward the hole. Removing mass shifts C away from it. Subtract hole inertia about O, then shift from O to the new C.

WATCH / Never assign the hole a mass M a²/R² using the remaining mass M.

Nonuniform rod: normalise first

Rod from x=0 to l; linear density lambda=cx. Need its mass, COM and transverse inertia.

$$M = cl^2/2, \quad c = 2M/l^2$$

$$x_C = \tfrac{1}{M}\int_0^l x(cx)dx = 2l/3$$

$$I_0 = \int_0^l x^2(cx)dx = Ml^2/2$$

$$I_C = I_0 - Mx_C^2 = Ml^2/18$$

WHY / The uniform-rod formulas do not apply. Fix the density constant from the total mass before integrating.

Check: C moves toward the denser end; I0 still has dimensions mass times length squared.

Radially varying disc density

A circular lamina has sigma(r)=c r^n, 0<r<R, n>-2. Here c is determined by mass, not arbitrary.

$$dm = 2\pi c r^{n+1} dr$$

$$M = \frac{2\pi c R^{n+2}}{n+2}$$

$$I_z = MR^2\frac{n+2}{n+4}$$

$$I_{\text{diameter}} = I_z/2$$

WHY / Divide the inertia integral by the mass integral; circular symmetry still equates central diameters.

n=0 gives a disc; n->infinity concentrates mass near the rim and approaches MR².

Cone & the right learning habit

A uniform solid cone of height h, base radius R has radius s=Rz/h at distance z from its vertex.

$$dm = \rho\pi(Rz/h)^2 dz$$

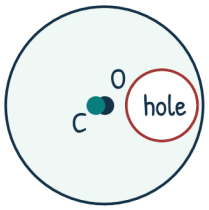
$$I_{\text{axis}} = \int_0^h \tfrac{1}{2}(Rz/h)^2 dm$$

$$I_{\text{axis}} = 3MR^2/10$$

WHY / Divide by M=rho pi R²h/3. This is the same disc-slicing idea as a sphere, not specialised rotational theory.

Prioritise rods, rings, discs, cylinders, spheres and axis shifts. For unfamiliar shapes know how to derive the needed result rather than memorising an unbounded catalogue.

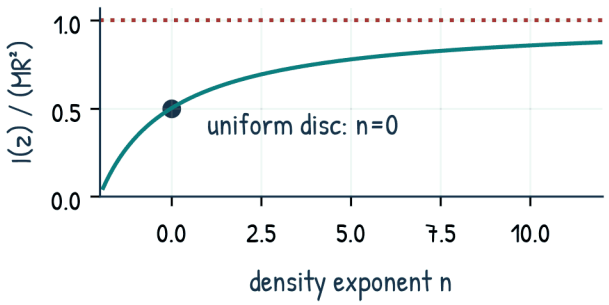
Off-centre hole: subtract mass AND its shifted inertia



same material density

C shifts away from the hole

Radial density r^n: more rim weighting increases axial MOI



REMEMBER / For a cut-out use the removed piece's original density and its own COM; shift all contributions to the requested axis.

Shift axes, do not guess

Parallel-axis shifts and perpendicular-axis splitting are different operations.

Parallel-axis theorem

$$I_A = I_C + Md^2$$

A and C are parallel axes; C passes through the centre of mass. d is their perpendicular separation.

WHY / Expand $(x+d)^2+y^2$ in the inertia integral. The cross term vanishes because integral x dm = 0 at C.

WATCH / For two arbitrary parallel axes A and B, subtract their separate COM shifts; do not just add $M(AB)^2$.

$$I_A = I_C + Md^2 + 2d\int x\,dm$$

$$\int x\,dm = 0 \quad \text{at C}$$

For fixed orientation, the parallel COM axis minimises I. Equality occurs only at d=0.

Which tangent axis?

For a disc, an axis through a rim point and parallel to the central normal gives:

$$I_A = \tfrac{1}{2}MR^2 + MR^2 = \tfrac{3}{2}MR^2$$

An axis tangent in the plane of the disc instead gives:

$$I_B = \tfrac{1}{4}MR^2 + MR^2 = \tfrac{5}{4}MR^2$$

WHY / Both are called "tangent" informally, but their directions differ. Draw the axis before substituting.

$$I_{\text{ring, tangent in plane}} = \tfrac{3}{2}MR^2$$

$$I_{\text{ring, rim normal}} = 2MR^2$$

A thin ring diameter has $I=MR^2/2$. Add MR^2 to the appropriate central-axis result.

Perpendicular-axis theorem

$$I_z = I_x + I_y \quad (z=0 \text{ lamina})$$

All three axes are mutually perpendicular and pass through the same point; x and y lie in the sheet.

WHY / For each element: distance² to z = x^2+y^2, which is the sum of the other two squared distances.

$$I_{\text{disc, diameter}} = MR^2/4$$

WATCH / Never use it for a solid sphere; mass away from the xy plane breaks the identity.

$$I_x = \int y^2 dm, \quad I_y = \int x^2 dm$$

$$I_z = \int (x^2+y^2) dm$$

The three axes must share a single origin, though that point need not be C.

A robust calculation route

1. Sketch the required axis. 2. Split into known pieces. 3. Shift each piece from its OWN COM. 4. Add signed inertias.

$$I_{\text{rectangle, normal}} = \frac{M(a^2+b^2)}{12}$$

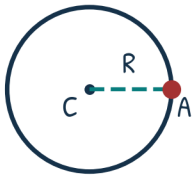
WHY / For a uniform rectangle, integrate x^2+y^2 separately; this also checks the rod limit as b tends to zero.

Check: l=2 m, M=3 kg rod; pivot 0.5 m from one end gives I = 1.75 kg m².

$$I_A - I_B = M(d_A^2 - d_B^2)$$

Here d(A) and d(B) are distances from the same parallel COM axis, not from each other.

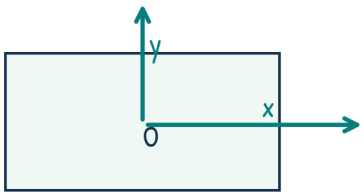
Disc rim axis parallel to the central normal



Both axes normal to disc plane

I(A) = $3MR^2/2$

Three common-origin axes; x and y in a lamina



z: out of plane at the same O

REMEMBER / Parallel-axis addition starts from a COM axis; perpendicular-axis splitting needs a planar lamina.

Torque, work & massive pulleys

For planar equations take anticlockwise as positive; write force components in the lab frame.

Turning effect of a force

$$\vec{\tau}_O = \vec{r}_{P/O} \times \vec{F}$$

$$|\tau_O| = F d_\perp$$

d is the shortest distance from O to the line of action. A couple has zero net force but nonzero, origin-independent torque.

WHY / Only the force component perpendicular to r turns the body about O.

WATCH / Torque is N m, not a scalar energy in joules; its sign follows the chosen axis.

$$\vec{\tau}_B = \vec{\tau}_A - \vec{r}_{B/A} \times \vec{F}_{\text{net}}$$

If net force is zero, net torque is origin-independent. A couple has magnitude F times the separation of its force lines.

Equations of motion

$$\sum \vec{F} = M\vec{a}_C$$

$$\sum \tau_{C,z} = I_C \alpha$$

The second equation is for planar motion. For a fixed spatial axis A, use its torque component and I about A.

WHY / Internal central forces cancel in the whole-body balance. Supports can exert force and torque.

WATCH / A general 3-D vector equation tau = I alpha is not valid. The fixed-axis scalar equation remains valid.

$$\sum F_x = Ma_{C,x}, \quad \sum F_y = Ma_{C,y}$$

Write the translation and rotation equations before inserting a no-slip or string constraint. Forces through C can still move C.

Energy and power

$$W = \int \tau_A \, d\theta = \Delta K$$

$$P = \tau_A \omega$$

For fixed-axis rotation with constant I: K = I omega² / 2. For free planar motion add M v(C)² / 2 to the spin energy.

WHY / Each tangential displacement is r dtheta, so summing F ds gives torque times angular displacement.

Check: I=10 kg m² at 1000 rpm stores 54.83 kJ.

$$P = \vec{F} \cdot \vec{v}_C + \tau_{C,z}\omega$$

For one applied force in planar motion, this equals its force dotted with the velocity of its actual point of application.

Atwood: include the pulley

Masses m2 > m1; pulley radius R, axial inertia I; light inextensible string does not slip; axle friction is negligible.

$$a = \frac{(m_2 - m_1)g}{m_1 + m_2 + I/R^2}$$

$$(T_2 - T_1)R = I\frac{a}{R}$$

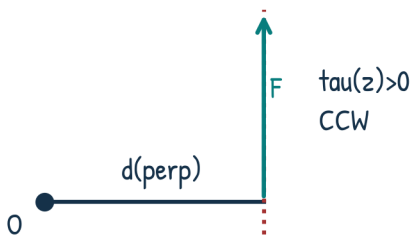
WHY / The pulley consumes part of the released potential energy. Tensions are equal only in the zero-inertia limit.

WATCH / No slip at a moving pulley rim does not mean tension does zero work on that pulley.

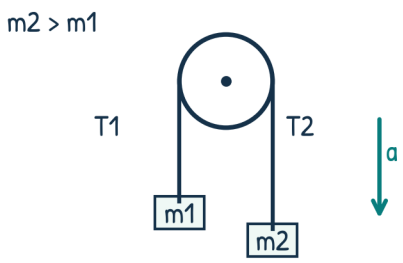
$$T_1 = m_1(g + a), \quad T_2 = m_2(g - a)$$

Add the mass equations to (T2-T1)=Ia/R²; the effective inertial mass is m1+m2+I/R².

Torque arm is perpendicular to the force line



Massive pulley: T2-T1 supplies the spin torque



REMEMBER / Use force balance for C and torque about C; a massive pulley usually has unequal string tensions.

Angular momentum: choose O

Specify system and origin before declaring conservation. Signed z components suffice for planar motion.

Spin plus orbital motion

$$\vec{L}_O = \vec{r}_C \times M\vec{v}_C + \vec{L}_C$$

$$L_{C,z} = I_C\omega \quad \text{(planar)}$$

A translating body can have nonzero L about O even with no spin. For a particle, L = r cross p.

WHY / Write each position as r(C)+rho and each velocity as v(C)+v(relative). COM cross terms sum to zero.

WATCH / L has units kg m²/s and changes when the origin changes.

$$L_z = mvb \quad \text{(particle)}$$

b is the signed perpendicular distance from O to the velocity line; a line through O gives L=0 even at large speed.

The conservation test

$$\vec{\tau}_{\text{ext, O}} = \frac{d\vec{L}_O}{dt}$$

This form uses absolute particle velocities and a fixed inertial origin O. Zero z torque conserves Lz, even if other components change.

WHY / For a central interaction, equal and opposite internal forces have no net torque.

WATCH / The CM is also a valid torque-rate origin. An arbitrary moving origin needs care; an instantaneous rest point is not enough.

$$\Delta L_z = \int \tau_{\text{ext},z}\, dt$$

Zero net force alone does not imply zero torque: a pure couple changes L without changing COM momentum.

Changing I at fixed L

$$I_1\omega_1 = I_2\omega_2$$

$$K = \frac{L_z^2}{2I}$$

Skater: I falls from 5 to 2 kg m²; omega rises from 2 to 5 rad/s; K rises from 10 to 25 J.

WHY / The extra 15 J comes from internal work, not external torque. This is a shape-changing system, not one permanently rigid body.

WATCH / Angular-momentum conservation is not energy conservation.

$$\Delta K = \frac{L_z^2}{2}(I_2^{-1} - I_1^{-1})$$

Conversely, spreading mass outward at fixed L decreases rotational KE; mechanical work must account for that change.

Two coaxial discs couple

$$\omega_f = \frac{I_1\omega_1 + I_2\omega_2}{I_1 + I_2}$$

$$Q = \frac{I_1 I_2(\omega_1 - \omega_2)^2}{2(I_1 + I_2)}$$

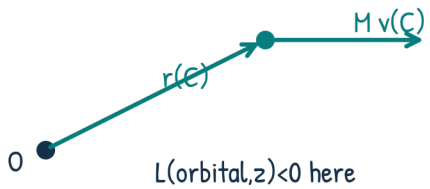
Use signed initial angular velocities; include both discs in the system and neglect external axial torque.

WHY / Friction is internal: it equalises spin while dissipating mechanical energy.

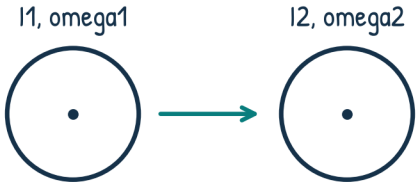
Check: I1=2, I2=3; omega1=10, omega2=-5 gives omega(f)=1 rad/s.

Derivation: conserve the sum I1 omega1+I2 omega2; insert the common final omega into Kf=(I1+I2) omega(f)²/2. Subtract Kf from Ki to obtain Q.

A translating mass can carry orbital angular momentum



Coaxial coupling: conserve total axial L, dissipate KE



REMEMBER / Zero external torque conserves angular momentum about that origin, not necessarily kinetic energy.

Rolling is a velocity constraint

Stationary rigid ground; wheel radius R. Here v is rightward and omega is clockwise-positive.

Match contact velocities

$$u = v - R\omega, \quad u = 0 \text{ (no slip)}$$

$$v = R\omega, \quad a = R\alpha$$

For constant R on a fixed straight surface, differentiate the signed constraint to get the acceleration relation.

WHY / Translation right and clockwise rotation give opposite velocities at the bottom.

WATCH / On a moving belt, match the contact velocity to the belt, not to zero.

$$v - R\omega = V_{\text{belt}}$$

For a moving horizontal belt use its signed lab speed V(belt). On a straight belt a-R alpha equals its acceleration.

The velocity field

$$v_{\text{top}} = 2v, \quad v_{\text{bottom}} = 0$$

The centre moves at v. At each side point the velocity has equal horizontal and vertical components, so speed = square root of 2 times v.

WHY / Add translation to rotation as vectors, not as magnitudes.

WATCH / At the contact material point, acceleration is upward with magnitude omega² R in straight pure rolling. Zero velocity does not mean zero acceleration.

$$\vec{a}_P = \vec{a}_C + \vec{\alpha} \times \vec{r}_{P/C}$$

$$-\omega^2 \vec{r}_{P/C}$$

At the bottom, the tangential terms cancel in pure rolling; the remaining term points upward.

Energy and contact-point I

$$K = \tfrac{1}{2}Mv^2 + \tfrac{1}{2}I_C\omega^2$$

$$K = \tfrac{1}{2}(I_C + MR^2)\omega^2$$

The second expression uses pure rolling and the instantaneous contact axis normal to the plane.

WHY / It is a correct instantaneous energy identity, obtained by substitution; no extra motion has been added.

WATCH / Do not add translational energy again when using the contact-axis form.

$$K = \tfrac{1}{2}M(1+\beta)v^2, \quad \beta = I_C/(MR^2)$$

Disc: K=3Mv²/4. Hoop: K=Mv². Solid sphere: K=7Mv²/10. All use v of C.

When the torque shortcut works

$$\tau_{P,z} = (I_C + MR^2)\alpha_z$$

For circular rolling on a fixed straight plane: choose the ground contact location P at that instant, and use a(C) tangent to the plane.

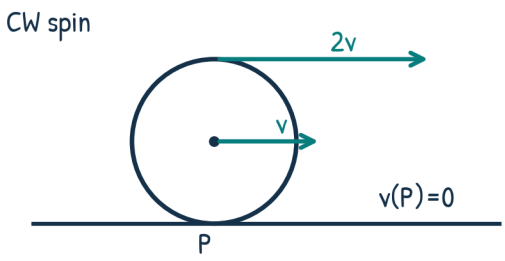
WHY / Torque about P = I(C) alpha + r(C/P) cross M a(C); the no-slip relation supplies the MR² term.

WATCH / This special geometry justifies the shortcut, not merely v(P)=0. On curved paths, derive the equation again.

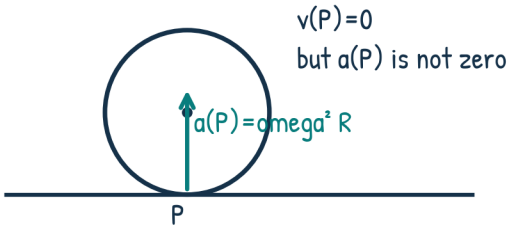
$$I_P = I_C + MR^2$$

In the scalar shortcut keep one signed rotation convention. Friction, weight-normal reaction and any force through P have zero moment about P.

Pure rolling on stationary straight ground



Pure rolling on stationary straight ground



REMEMBER / The material point at contact is instantaneously at rest; it is not a permanent fixed pivot.

Inclines: friction sets the regime

A symmetric body rolls freely from rest: I(C) = beta M R². Ignore rolling resistance and air drag.

Force plus torque

$$Mg\sin\theta - f = Ma$$

$$fR = I_C\alpha, \quad a = R\alpha$$

$$a = \frac{g\sin\theta}{1+\beta}$$

WHY / Gravity accelerates C while uphill static friction provides the torque needed to spin the body.

WATCH / These formulas assume no driving torque and no other tangential force. Do not reuse them for a pulled spool.

$$\beta = I_C/(MR^2), \quad f = \beta Ma$$

Eliminate f from the two balances. For uphill motion the acceleration still points downhill; friction is still uphill in this force-free case.

Static is an inequality

$$f = \frac{\beta}{1+\beta}Mg\sin\theta$$

$$N = Mg\cos\theta$$

$$\mu_s \geq \frac{\beta}{1+\beta}\tan\theta$$

WHY / Static friction takes the required value, not automatically mu(s) N. Equality is the threshold of slipping.

Sphere at 30°, g=10: a=25/7 m/s²; minimum mu(s)=0.165.

$$\tan\theta \leq \mu_s(1+\beta)/\beta$$

If this fails from release, use kinetic friction and separate a from R alpha; no-slip energy formulas cannot be imposed.

Energy tells speed and height

$$v^2 = \frac{2gh}{1+\beta}$$

$$K_{\mathrm{rot}}/K = \frac{\beta}{1+\beta}$$

Starting from rest through vertical drop h, gravity supplies both types of kinetic energy. Static friction has zero total power on fixed ground.

WHY / Friction's translational and rotational powers cancel; it can redistribute the two energies.

A rolling hoop climbing at 5 m/s on 30°, g=10, rises 2.5 m: path length 5 m.

$$h_{\mathrm{stop}} = \frac{(1+\beta)v_0^2}{2g}$$

For pure rolling up a straight incline, path distance is h(stop)/sin(theta). Stop and reverse if the constraint remains valid.

Race and smooth-plane limits

Fast to slow: solid sphere, solid cylinder, thin spherical shell, hoop. The ratio beta, not mass alone, decides.

$$\beta: \quad 2/5,\ 1/2,\ 2/3,\ 1$$

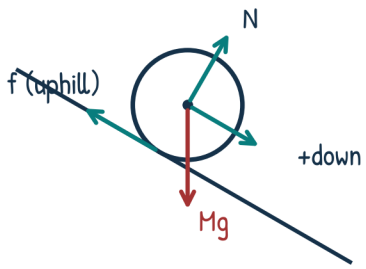
WHY / Smaller beta needs a smaller rotational share for a given translational speed.

WATCH / A frictionless block is faster than these rollers. On a perfectly smooth plane, an initially nonspinning symmetric ball slides without spinning up.

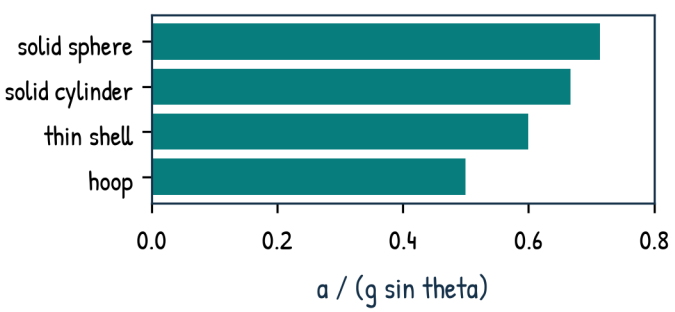
$$v_{\mathrm{disc}}/v_{\mathrm{hoop}} = \sqrt{4/3}$$

At the same vertical drop, a smooth block has v=sqrt(2gh). A rough sliding block is not necessarily fastest.

Free rolling: static friction points up the incline



Rolling race: beta alone sets normalised acceleration



REMEMBER / First solve for the static friction required; only then check whether the surface can supply it.

Slipping: follow the contact speed

Horizontal fixed ground; constant kinetic coefficient mu(k); +x right, clockwise omega positive.

Separate translation and spin

$$u = v - R\omega, \quad s = \operatorname{sgn}(u_0)$$

$$\dot{v} = -s\mu_k g$$

$$R\dot{\omega} = s\mu_k g/\beta$$

beta = I(C)/(M R²) > 0. These equations apply until u reaches zero, with constant normal force Mg.

WHY / The same friction slows contact slip through translation and rotation simultaneously.

WATCH / Do not impose v = R omega during this phase.

$$\dot{u} = -s\mu_k g(1 + 1/\beta)$$

Friction direction stays fixed until u first vanishes. If v crosses zero earlier, that alone is not a reason to reverse friction.

Transition and final speed

$$t_* = \frac{|v_0 - R\omega_0|}{\mu_k g(1 + 1/\beta)}$$

$$v_* = \frac{v_0 + \beta R\omega_0}{1 + \beta}$$

At t*, R omega* = v*. With no further horizontal forces or torques, friction becomes zero and pure rolling persists.

WHY / The combination v + beta R omega is constant during slip.

WATCH / If u0=0 there is no slipping stage. If mu(k)=0 and u0 is nonzero, no finite transition occurs.

$$v + \beta R\omega = v_0 + \beta R\omega_0$$

Derivation: add the two acceleration equations with the second multiplied by beta. Set the final slip to zero.

Where the energy went

$$\dot{K} = -\mu_k Mg|u|$$

$$Q = \frac{M\beta}{2(1+\beta)}u_0^2$$

Q is the loss of mechanical energy over the complete sliding stage. Total energy including heat is conserved.

WHY / Power at the slipping contact uses its velocity u, not merely the velocity of C.

WATCH / Sufficient backspin can stop or reverse the ball: v* can be positive, zero or negative.

$$Q = \mu_k Mg\frac{|u_0|t_*}{2}$$

The slip-speed graph is a triangle. Its area is relative sliding distance, not COM displacement.

Corrected bowling-ball check

Solid sphere: beta=2/5, R=0.1 m, v0=10 m/s, omega0=0, mu(k)=0.3, g=10 m/s².

$$\alpha = 75 \text{ rad/s}^2, \quad t_* = 20/21 \text{ s}$$

$$v_* = 50/7 \text{ m/s}, \quad Q = Mv_0^2/7$$

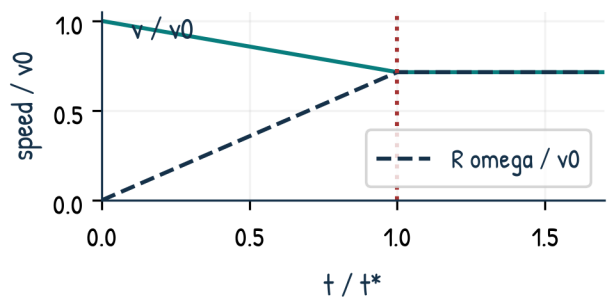
WHY / The source E4 misses a factor of ten in alpha; P29 and OL1 give the consistent transition.

For v0=10 and R omega0=-5 with mu(k)=0.2: t*=15/7 s; v*=40/7 m/s.

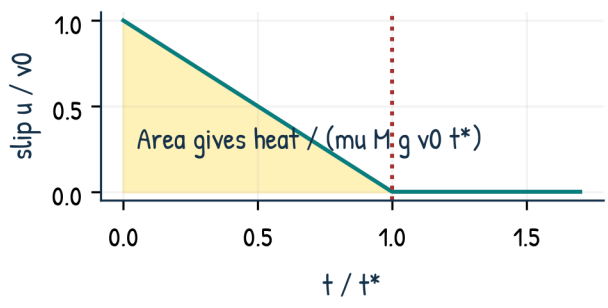
$$x_* = v_0 t_* - \tfrac{1}{2}s\mu_k g t_*^2$$

For the no-spin numerical case x*=400/49 m. This is a COM displacement measured from first contact.

No-spin solid sphere: slopes change at t*



Slip vanishes continuously; dissipative power then ceases



REMEMBER / Kinetic friction opposes u = v - R omega, which need not have the same sign as the centre velocity.

Driven rollers & spool reversal

Horizontal ground; +x right, angular quantities clockwise-positive. Assume no slip, then verify it.

One useful master equation

Apply horizontal force F to the right along a line height h above C. Let friction f be positive rightward.

$$F+f=Ma, \quad Fh-fR=I_C\alpha$$

$$a=\frac{F(1+h/R)}{M(1+\beta)}$$

$$f=\frac{F(h/R-\beta)}{1+\beta}$$

WHY / Eliminating friction with a = R alpha exposes the line-of-action lever arm. h is signed.

$$|f|\le\mu_s Mg$$

At h=beta R no friction is needed: the applied force supplies translation and spin in exactly the rolling ratio.

Centre pull versus top pull

At h=0: a=F/[M(1+beta)], with friction backward. At h=R for a disc (beta=1/2):

$$a=4F/(3M), \quad f=+F/3$$

WHY / A top pull supplies more spin torque than needed without friction. Forward friction adds translation and reduces net clockwise torque.

WATCH / Source E9's acceleration is right, but its claimed backward friction is wrong.

In both cases require |f| <= mu(s) Mg.

$$f_{h=0}=-\frac{\beta F}{1+\beta}$$

Disc centre-pull: a=2F/(3M), f=-F/3. Contrasting it with a top pull is a quick sign check.

Spool: state the winding

String leaves the INNER hub on the lower-right tangent and is pulled up-right at angle phi. Outer radius R, hub radius r, axial inertia I.

$$Ma=F\cos\phi+f$$

$$I_C\alpha=-Fr-fR$$

$$a=\frac{F(\cos\phi-r/R)}{M(1+\beta)}$$

WHY / About contact, the string's clockwise lever arm is R cos(phi) - r. Reversing the winding reverses the hub torque.

$$a_{\text{upper}}=\frac{F(\cos\phi+r/R)}{M(1+\beta)}$$

For an upper-hub tangent the string torque is clockwise, so the reversal criterion differs. Draw the tangent explicitly.

Direction is geometric

$$\cos\phi_c=r/R$$

Below phi(c), a is toward the puller; above phi(c), it is away. This is acceleration from rest, not an instant velocity reversal.

$$N=Mg-F\sin\phi, \quad f=Ma-F\cos\phi$$

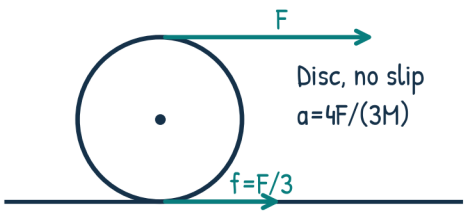
WHY / mu(s) decides if the assumed rolling solution is possible; it does not set the critical angle.

WATCH / Need N>0 and |f| <= mu(s) N. Source P35 lacks F and I, so its rolling feasibility cannot be decided numerically.

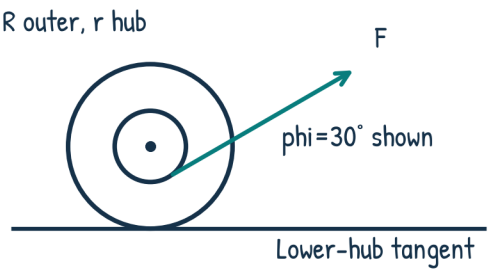
$$\phi_c=\cos^{-1}(r/R)$$

For r/R=1/4, phi(c)=75.52°. At the critical angle a=0 and f=-F cos(phi), provided static contact can support it.

Top-pulled disc: friction is forward, not backward



Specified winding: hub torque from F is counterclockwise



REMEMBER / A static friction direction must be solved, not guessed from the direction of translation.

Unwinding strings & energy

A fixed upper string end; vertical descent; string taut, light and inextensible; no slip at hub.

A yo-yo is a core application

Let a be downward acceleration, T upward tension, I the axial inertia about C. Choose positive spin in the unwinding direction.

$$Mg - T = Ma, \quad Tr = I\alpha$$

$$a = r\alpha$$

WHY / The string contact has zero velocity in the lab when the top end is fixed. That is the geometrical constraint.

WATCH / If the top end moves, the constraint and tension power change.

$$v = r\omega$$

The fixed string cancels the velocity of the hub contact; it does not make the body's centre stationary.

Solve before substituting I

$$a = \frac{Mgr^2}{Mr^2 + I}$$

$$T = \frac{MgI}{Mr^2 + I}$$

$$v^2 = \frac{2Mgh}{M + I/r^2}$$

WHY / Tension partly cancels gravity while providing angular acceleration. In the fixed-string idealisation it does no total work.

For release from rest through height h, either use $v^2=2ah$ or conservation of mechanical energy.

$$Mgr = (Mr^2 + I)\alpha$$

This is the combined equation with a=r alpha. It also checks the acceleration dimensions without assuming a solid disc.

Solid-disc model

$$I = \tfrac{1}{2}MR^2$$

$$a = \frac{2gr^2}{2r^2 + R^2}$$

$$T = \frac{MgR^2}{2r^2 + R^2}$$

WHY / For r=R: a=2g/3 and T=Mg/3, so a+T/M=g as required.

WATCH / Source P31 has an extra factor 1/2 in T. Small r means a small, but tension tends to Mg, not to zero.

$$r \ll R: \quad a \simeq 2g(r/R)^2$$

As I tends to zero at fixed r, a tends to g and T tends to zero. As I grows large, a tends to zero and T tends to Mg.

An energy-accounting habit

Choose the whole body as system. Use translational plus rotational KE and include gravity through potential energy.

$$Mgh = \tfrac{1}{2}Mv^2 + \tfrac{1}{2}I(v/r)^2$$

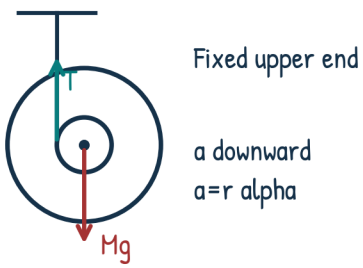
WHY / Writing the constraint inside the energy formula makes the effective inertia M+I/r² visible.

WATCH / A real yo-yo has finite string length, layer-dependent hub radius, bearing losses and possible rebound. This sheet models only ideal unwinding.

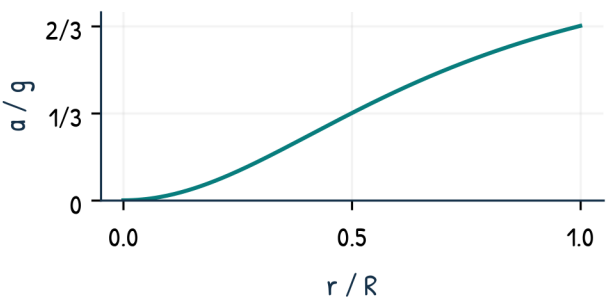
$$Tr\omega - Tv = 0$$

Tension removes translational power and supplies the same rotational power. Gravity supplies the net increase in total KE.

Yo-yo unwinds about hub radius r



Solid-disc yo-yo: r/R 0 gives a 0



REMEMBER / The unwinding radius is the hub radius r, not necessarily the outer body radius R.

Impact first, swing second

A short embedding impact, a fixed frictionless hinge O, and negligible gravitational angular impulse.

The right system and origin

Include bullet m and rod M, length l. The rod is initially at rest. Bullet speed v is perpendicular to the rod and hits distance d from O.

$$I_f = Ml^2/3 + md^2$$

$$mvd = I_f\omega_+$$

WHY / The hinge impulse can be large, but its torque about O is zero. Gravity's angular impulse is negligible over the short impact.

WATCH / Total linear momentum is generally not conserved: the hinge supplies external impulse.

$$\omega_+ = \frac{mvd}{Ml^2/3 + md^2}$$

If the bullet velocity makes angle gamma with the rod, replace vd by vd sin(gamma); use the signed moment arm.

Energy lost in embedding

$$K_+ = \frac{m^2v^2d^2}{2(I_0 + md^2)}$$

$$Q = \frac{1}{2}mv^2\frac{I_0}{I_0 + md^2}$$

Here I0 = M l² / 3. Q decreases as d increases for fixed bullet speed and rod.

WHY / A hit farther out transfers a larger fraction of the initial energy into rotation.

WATCH / P28(c) reverses this: maximum loss is at d=0 (or the smallest allowed d), not at the far end.

At d->0, final spin and final KE vanish; Q tends to the whole incident bullet KE. Do not maximise omega to maximise the loss.

Subsequent upward swing

$$\tfrac{1}{2}I_f\omega_+^2 = (Mgl/2 + mgd)(1 - \cos\theta)$$

The rod began vertically downward; theta is its angular displacement from that position. Each mass has its own height change.

WHY / The impact has ended. Now gravity is conservative, and a stationary ideal hinge does no work.

WATCH / A turning angle exists only if the energy is at most 2(Mg l/2+mgd). Above the barrier, the ideal hinged system can pass over the top.

$$\cos\theta_{\max} = 1 - \frac{I_f\omega_+^2}{2g(Ml/2 + md)}$$

This gives a turning point only when the right-hand side lies between -1 and 1.

Numbers and reusable impulse

m=0.01 kg, v=400 m/s, M=1 kg, l=1 m, d=0.5 m:

$$I_f = 0.335833\ \text{kg m}^2$$

$$\omega_+ = 5.9553\ \text{rad/s}$$

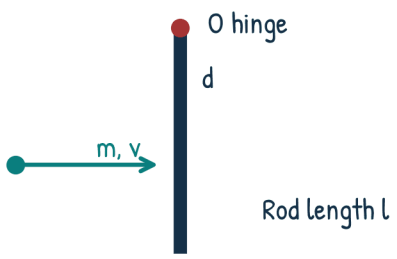
$$\Delta L_O = \int \tau_{\text{ext, O}}\, dt$$

WHY / If the impact force is not perpendicular, replace mvd by mv times the perpendicular impact parameter.

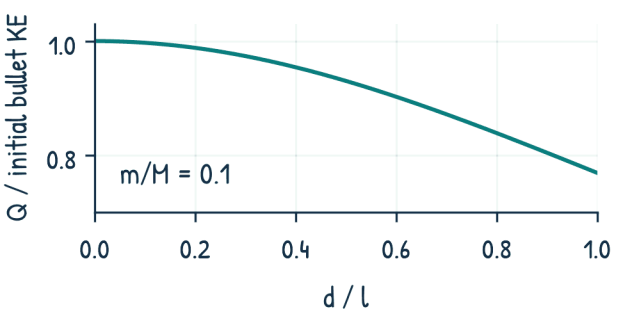
WATCH / A frictional hinge, off-axis impact or finite collision time may invalidate the zero-angular-impulse approximation.

After impact the bullet and rod are one composite body. Its new inertia includes the embedded point mass; using the bare rod inertia misses angular momentum.

Bullet impact: take angular impulse about fixed hinge O



Embedding loss decreases with distance from the hinge



REMEMBER / Across impact use angular impulse; after impact use energy if the hinge is ideal and no dissipation acts.

Impulses: free bodies & hinges

During a short impact ignore gravity impulse; smooth supports give no horizontal impulse. Keep all spin signs consistent.

Impulse on an initially resting body

Impulse magnitude J acts horizontally with perpendicular lever arm b from the target COM. Mass M, planar COM inertia I.

$$V_C = J/M, \quad |\omega| = Jb/I$$

$$K = \tfrac{J^2}{2}(1/M + b^2/I)$$

WHY / Integrate F=Ma for translation and torque=I alpha about C for spin. Add their kinetic energies.

The contact-point speed component along J is J(1/M+b²/I); define an effective mass by the reciprocal of this bracket.

WATCH / A force line through C (b=0) gives pure translation, not spin.

A particle hits the free target

Incoming particle mass m and speed v. The target initially rests. The impact impulse is along the incident line; e is normal restitution.

$$D = 1/m + 1/M + b^2/I$$

$$J = (1+e)v/D$$

$$v_{m,+} = v - J/m$$

WHY / The target's contact speed is J(1/M+b²/I). Restitution sets target-contact speed minus outgoing particle speed equal to e v.

$$Q = (1-e^2)v^2/(2D)$$

WATCH / e=0 means zero relative separation speed in the impact direction; it does not by itself enforce permanent sticking.

Hinged rod: support impulse

A resting rod is hinged at O. Its COM is distance d from O; a perpendicular impulse J strikes distance b from O. I(O) is the hinge inertia.

$$\omega_+ = Jb/I_O, \quad V_C = d\omega_+$$

$$J_H = MV_C - J$$

$$J_H = J(Mdb/I_O - 1)$$

WHY / Angular impulse about the hinge eliminates its unknown impulse; linear impulse then determines the hinge reaction.

J(H) is signed along the incoming impulse. It can oppose it, vanish, or act in the same direction.

Centre of percussion

$$b_{\text{cp}} = I_O/(Md)$$

At this strike distance, J(H)=0: no transverse shock is transmitted to the ideal hinge.

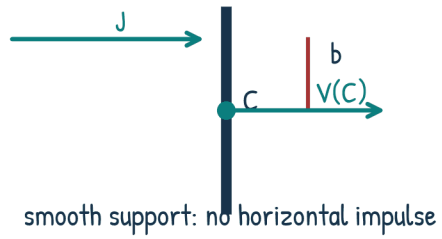
$$\text{uniform rod}: \quad b_{\text{cp}} = 2l/3$$

WHY / Insert I(O)=Ml²/3 and d=l/2. This is the same length I(O)/(Md) that appears in the physical pendulum.

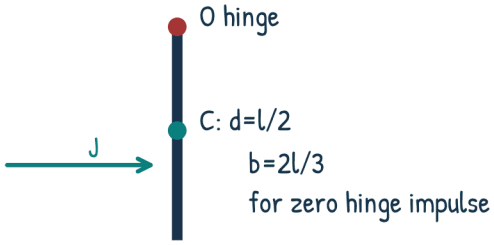
Conditions: initially stationary rigid body, fixed hinge, short transverse impulse, no other transverse impulses.

WATCH / A free rod and a hinged rod are different systems. Do not conserve their linear momentum in the same way.

Impulse above C: translation right and clockwise spin



Uniform end-hinged rod: centre of percussion at 2l/3



REMEMBER / A free body translates and spins after an eccentric impulse. A hinge changes which momentum equation is useful.

Equilibrium, slipping & tipping

For static balance require BOTH zero net force and zero net torque; contact reactions are constrained.

Ladder: draw every force

Uniform ladder length l at angle theta to horizontal. Wall smooth, floor rough. Weight Mg acts at its midpoint.

$$N_g = Mg, \quad f = N_w$$

$$N_w l \sin\theta = Mg(l/2)\cos\theta$$

$$\mu_s \geq \tfrac{1}{2}\cot\theta$$

WHY / Take torques about the base to eliminate both ground reactions.

WATCH / Equality holds only at impending slip. This ideal ladder has point contacts, not a finite block base.

$$f \leq \mu_s N_g, \quad N_w, N_g \geq 0$$

An extra person or load adds its own moment about the base. Recompute rather than reuse the uniform-ladder threshold.

Block on an incline

Half-width b along the plane; COM height h measured normal to it. With no other force, the weight projection must remain inside the base.

$$\tan\theta_{\text{tip}} = b/h$$

$$\tan\theta_{\text{slide}} = \mu_s$$

WHY / At tipping, the ground normal resultant reaches the downhill edge and the weight has zero moment about that edge.

WATCH / Whichever critical angle is smaller occurs first. The tipping result presumes sliding has not already begun.

$$F_{\text{tip}} = Mgb/y, \quad F_{\text{slide}} = \mu_s Mg$$

For a level-floor block pushed horizontally at height y, compare these thresholds; b is half-width of the base.

Accelerating supports

In the frame of a support accelerating horizontally right with A, include pseudo-force M A leftward at C.

$$\vec{g}_{\text{eff}} = \vec{g} - \vec{A}$$

The effective-weight line through C must intersect the support footprint for quasistatic balance.

WHY / This is simply torque balance written in a translating non-inertial frame.

WATCH / Use this static construction only when the body remains at rest relative to that frame.

$$A_{\text{tip}} = gb/h$$

For the same block carried by a horizontal support, the effective-weight line reaches the rear edge at this acceleration, if it has not slipped first.

Overhang: COM recursion

$$\delta_n = \frac{l}{2n}$$

$$D_N = \frac{l}{2}\sum_{n=1}^{N}\frac{1}{n}$$

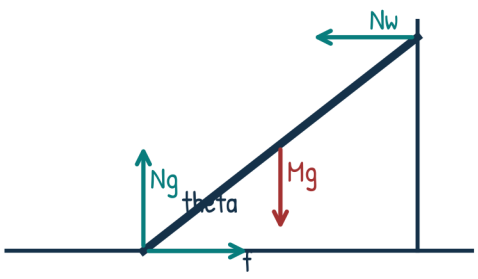
Identical horizontal blocks, ideal rigid contacts; delta(n) is the offset between block n and the support below the top n blocks.

WHY / Place each upper stack's COM exactly above the next support edge. Harmonic growth is logarithmic, not linear.

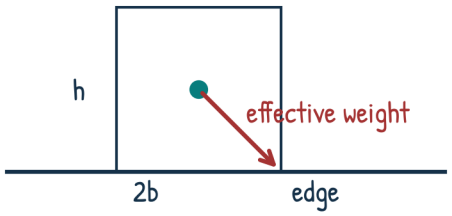
Checks: D4=25l/24; D10=1.46448l; D(10^6)=7.19636l. Exact balance is marginal.

For the top n blocks their combined COM must lie above the next support. Adding one block shifts that COM; balancing this shift gives l/(2n).

Smooth wall, rough floor: four forces on the ladder



Tip threshold: effective-weight line reaches a support edge



REMEMBER / A body can lose contact, start sliding or start toppling; check which inequality fails first.

Energy, reactions & curved rolling

Harder JEE applications still use the same core laws. Energy gives speed; force balance tests contact.

A rod falls about a fixed hinge

Uniform rod length l, mass M, released horizontally from rest. theta is its downward angle from the initial horizontal.

$$Mg(l/2)\sin\theta = \tfrac{1}{2}(Ml^2/3)\omega^2$$

$$\omega^2 = (3g/l)\sin\theta$$

$$\alpha = (3g/2l)\cos\theta$$

WHY / Take gravitational torque about the hinge. Energy supplies omega; torque supplies alpha.

At the lowest point omega=sqrt(3g/l), alpha=0, but the COM still has inward centripetal acceleration.

Hinge reaction is not Mg

At release the rod is horizontal; C accelerates downward with magnitude alpha l/2=3g/4.

$$H_y - Mg = -3Mg/4 \;\Rightarrow\; H_y = Mg/4$$

At the lowest point C accelerates upward at omega²l/2=3g/2.

$$H_y - Mg = 3Mg/2 \;\Rightarrow\; H_y = 5Mg/2$$

WHY / The fixed hinge does no work, but it does supply force and impulse. "No work" does not mean "no force".

WATCH / Between these positions use BOTH tangential and radial COM acceleration before resolving the hinge reaction.

Rolling inside a vertical loop

Track inner radius R; body radius r; q=R-r is the COM path radius. beta=I(C)/(M r²). theta is measured from the bottom.

$$v^2 = 2g[h - q(1-\cos\theta)]/(1+\beta)$$

$$N = M(v^2/q + g\cos\theta)$$

WHY / Energy uses the COM height. Radially inward: N-Mg cos(theta)=Mv²/q. At the top this becomes N+Mg=Mv²/q.

h is the initial COM height above its lowest path point. Require N>=0 along the motion.

WATCH / Do not use the track radius R in the COM centripetal term.

Minimum height: a conditional result

$$v_{\text{top}}^2 \geq gq$$

$$h \geq \frac{5+\beta}{2}q$$

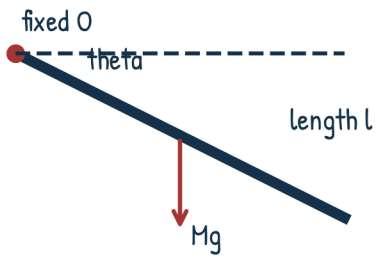
Derive by imposing N(top)>=0 in the energy equation. beta=0 recovers the particle loop result 5q/2.

WHY / This is the ideal-constraint energy/contact lower bound, not a guarantee for a real rough roller.

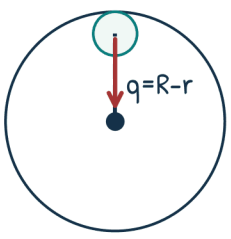
$$|f| = \frac{\beta}{1+\beta}Mg|\sin\theta|$$

WATCH / Check |f|<=mu(s)N everywhere. Near marginal top contact, finite friction may be insufficient even if the top-speed condition holds. A higher release height can be needed.

Hinged rod falls through theta below its initial horizontal



Inside loop: the COM follows radius R-r, not R



N, Mg: inward

at top: v² >= g(R-r)
check friction
elsewhere too

REMEMBER / Energy conservation alone cannot establish continued contact, pure rolling, or an admissible support force.

Rigid-body oscillations: SHM link

This is an application of rotational mechanics to the JEE oscillations chapter, not specialised top theory.

Physical pendulum

A fixed horizontal pivot axis is d above C in stable equilibrium. I(O) is about that pivot.

$$I_O\ddot{\theta} = -Mgd\sin\theta$$

$$T = 2\pi\sqrt{I_O/(Mgd)}$$

WHY / For |theta| much smaller than 1 rad, sin(theta) is approximately theta; the torque then has the SHM form.

WATCH / The period formula is small-amplitude only. At d=0 there is no gravitational restoring torque.

$$I_O = I_C + Md^2$$

Use the pivot axis, not the COM axis, in the period. The restoring sign is negative about downward equilibrium.

Equivalent length and optimum

$$l_{\text{eq}} = \frac{I_O}{Md} = d + \frac{k_C^2}{d}$$

$$d_{\text{opt}} = k_C, \quad l_{\text{eq, min}} = 2k_C$$

Use I(O)=M(k(C)²+d²). The optimum is valid only if that pivot position is physically allowed.

WHY / Differentiate d+k²/d: the two terms balance at d=k.

Disc: d=R/sqrt(2); end-pivot rod: T=2 pi sqrt(2l/3g).

$$T_{\text{rod, end}} = 2\pi\sqrt{2l/(3g)}$$

$$T_{\text{disc, rim}} = 2\pi\sqrt{3R/(2g)}$$

Memorise the standard results, but know which d and I produce each one.

Cylinder in a circular bowl

Track radius R, roller radius r<R, beta=I(C)/(M r²). Let q=R-r; theta measures C's displacement from the bottom.

$$v_C = q\dot{\theta}, \quad |\omega| = v_C/r$$

$$K = \tfrac{1}{2}M(1+\beta)q^2\dot{\theta}^2$$

WHY / No slip on a fixed track gives rolling spin from arc travel of C. This is not spin equal to theta-dot.

WATCH / The relevant COM path radius is R-r, not R.

$$U = Mgq(1-\cos\theta)$$

Energy is K+U constant. Differentiate it to get the exact equation, then linearise; there is no small-angle step in the kinetic energy.

Small bowl oscillations

$$\ddot{\theta} = -\frac{g\sin\theta}{q(1+\beta)}$$

$$T = 2\pi\sqrt{q(1+\beta)/g}$$

For a solid cylinder beta=1/2, T is sqrt(3/2) times the simple-pendulum period for length q.

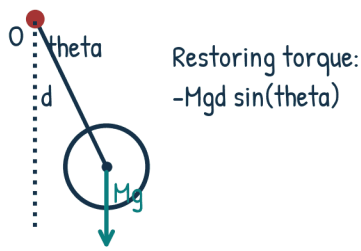
WHY / The exact restoring term is sine; only small oscillations are SHM. Rotation increases the effective inertia.

WATCH / Require contact and enough static friction throughout. The q->0 limit is geometrically singular, not a real zero-period oscillator.

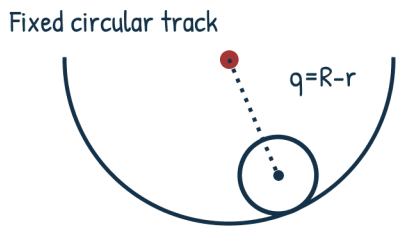
$$\frac{T}{T_{\text{simple}}} = \sqrt{1+\beta}$$

As beta->0 recover a point mass on a circular path. The effective inertia, not extra mass, lengthens the period.

Physical pendulum: measure theta from downward vertical



The roller centre follows a circle of radius R-r



REMEMBER / First identify the coordinate and inertia, then linearise the restoring torque for small oscillations.

Sliding rod & final JEE checks

A harder core application stays in JEE: use constraints, energy and a contact-force test.

Geometry before equations

Uniform rod length l on a smooth floor and wall. Theta is its angle ABOVE THE FLOOR, released from rest at 0<theta0<90°.

$$x_C = \tfrac{l}{2}\cos\theta, \quad y_C = \tfrac{l}{2}\sin\theta$$

$$K = \tfrac{Ml^2}{6}\dot{\theta}^2$$

WHY / C moves on a circle of radius l/2 while both contacts persist. Add spin inertia Ml²/12 to the translational term.

WATCH / A smooth floor does not pin the bottom. Never assume rotation about a fixed bottom end.

$$v_C^2 = (l^2/4)\dot{\theta}^2$$

While both contacts persist, the instantaneous centre is the intersection of their velocity normals; it is not the bottom endpoint.

Energy and angular acceleration

$$\dot{\theta}^2 = \tfrac{3g}{l}(\sin\theta_0 - \sin\theta)$$

$$\ddot{\theta} = -\tfrac{3g}{2l}\cos\theta$$

WHY / The potential drop is Mg l (sin(theta0)-sin(theta))/2. Differentiating energy gives the equation away from the initial rest instant; it extends continuously there.

The rod falls with theta decreasing; theta-dot is negative, while its squared value above stays positive.

$$\tfrac{Ml^2}{6}\dot{\theta}^2 = \tfrac{Mgl}{2}(\sin\theta_0 - \sin\theta)$$

The effective rotational inertia in this one-coordinate energy is Ml²/3, but the body is not rotating about a fixed floor point.

Contact loss: correct angle

$$N_w = M\ddot{x}_C$$

$$N_w = \tfrac{3Mg}{4}\cos\theta(3\sin\theta - 2\sin\theta_0)$$

$$\sin\theta_* = \tfrac{2}{3}\sin\theta_0$$

$$v_{C,*} = \tfrac{1}{2}\sqrt{gl\sin\theta_0}$$

WHY / A wall can push but cannot pull: set Nw=0 before the formula would predict a negative normal.

WATCH / Source OL3/P33 use cosine with a floor-angle energy convention; that is inconsistent.

$$N_{g,*} = Mg/4$$

The floor still pushes at detachment. Afterwards the wall force is zero and the two-contact circular-COM constraint must be dropped.

Before submitting a solution

Name the system, axis and positive direction. Match every I to its axis. Check no slip, contact, tension and friction limits.

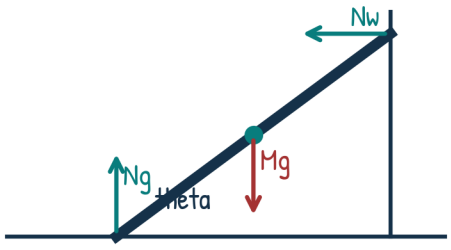
Separate impact, sliding, rolling and free-flight phases. Identify when energy, linear momentum or angular momentum is actually conserved.

WHY / Most "rotation traps" are an unspoken constraint or origin choice, not difficult algebra.

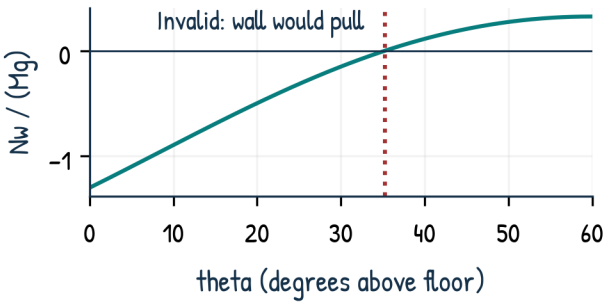
WATCH / Exactly vertical release at perfect rest is a degenerate ideal solution. Use a small perturbation to discuss falling; do not substitute it blindly into a claimed motion.

Fast checks: dimensions of I; moment arm perpendicular to force; torque sign; beta consistent with the body; mu(s) inequality; normal and tension nonnegative; physically allowed limiting case.

Smooth wall and floor: bottom can slide horizontally



theta0=60°: detachment at 35.264° (motion right to left)



REMEMBER / Keep the two-contact solution only until the wall normal first reaches zero; then change the constraints.

Beyond scalar inertia

Specialised extension: general 3-D rotation. The JEE sequence needs only planar and fixed-axis components.

Tensor definition

$$I_{ij} = \int (r^2 \delta_{ij} - x_i x_j)\, dm$$

$$\vec{L}_C = \mathbf{I}_C \vec{\omega}$$

$$K_{\text{spin}} = \tfrac{1}{2} \vec{\omega} \cdot \mathbf{I}_C \vec{\omega}$$

Coordinates are relative to C; delta(ij) is 1 for equal indices, 0 otherwise. I is symmetric and has orthogonal principal axes.

WHY / Each mass contributes r cross (omega cross r). Collecting coefficients gives this linear map.

Dumbbell: a clean counterexample

Two point masses m at ±a n, where n is a unit vector along a massless rod.

$$\mathbf{I} = 2ma^2(\mathbf{1} - \vec{n}\vec{n}^{\,T})$$

$$\vec{L} = 2ma^2[\vec{\omega} - (\vec{\omega} \cdot \vec{n})\vec{n}]$$

WHY / A rotation component along the zero-thickness rod contributes no angular momentum. L is perpendicular to n.

WATCH / OL8's component expression uses a non-orthogonal direction incorrectly. For generic oblique spin, L is not parallel to omega.

Body frame versus lab frame

$$\vec{\tau}_C = \left(\frac{d\vec{L}_C}{dt}\right)_{\text{body}} + \vec{\omega} \times \vec{L}_C$$

In principal axes the body-frame inertia matrix is diagonal. The extra cross term accounts for rotation of the basis itself.

WHY / A vector fixed in the body can still change direction in the laboratory.

WATCH / Do not differentiate body-frame components as if their basis vectors were fixed in space.

An arbitrary moving origin

$$\vec{L}_A = \sum_i (\vec{r}_i - \vec{r}_A) \times m_i \vec{v}_i$$

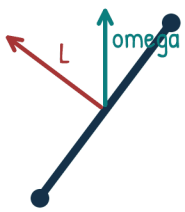
$$\frac{d\vec{L}_A}{dt} = \vec{\tau}_A - \vec{v}_A \times M\vec{v}_C$$

These velocities are absolute inertial velocities. For A=C the cross term vanishes. For relative velocities, the corresponding rate formula instead involves a(A).

WHY / Differentiating r(i)-r(A) generates the transport term.

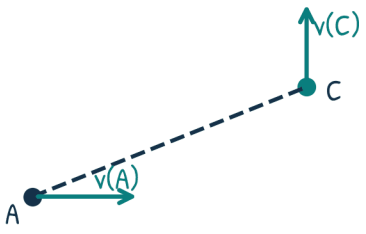
WATCH / A moving axis is not automatically a conserved-L origin just because its external torque vanishes.

Oblique dumbbell: subtract the component along the rod



L dot n = 0

Moving origins require the angular-momentum transport term



REMEMBER / The scalar I about an axis remains useful even when the full angular-momentum vector is not parallel to omega.

Gyroscopes: torque turns L

Specialised symmetric-top dynamics; separate spin, precession and nutation.

Geometry removes sin(theta)

Fixed pivot O; C at distance d along symmetry axis n. L is approximately I3 omega3 n for dominant spin.

$$|\tau| = Mgd\sin\theta$$

$$|\dot{\vec{L}}| = \Omega L\sin\theta$$

$$\Omega \simeq \frac{Mgd}{I_3\omega_3}$$

WHY / Both torque and the precession-circle radius contain sin(theta). They cancel at nonzero tilt.

WATCH / Use tau/(L sin(theta)), not tau/L for the precession rate around vertical.

When slow precession is safe

A nearly symmetric top, spin angular momentum dominant, slow steady precession and small nutation amplitude. Finite tilt is allowed.

$$I_\perp \Omega \ll |I_3\omega_3|$$

I(perp) is the transverse inertia about the pivot, not the COM. Initial conditions can excite nutation even at modest tilt.

WHY / Rapid spin gives a large L, so a given torque turns it slowly.

WATCH / As spin tends to zero the approximation fails; the predicted divergence does not mean an instantaneous fall.

Steady-precession branches

$$I_\perp \Omega^2\cos\theta - I_3\omega_3\Omega + Mgd = 0$$

For a heavy symmetric top in regular precession, theta is constant; omega3 is the actual angular velocity component along n.

WHY / The quadratic retains transverse angular momentum. The small root reduces to slow precession when spin dominates.

WATCH / Reality of roots and initial conditions matter. This is not the same as stability of an upright sleeping top.

Upright "sleeping" top

Ideal heavy symmetric top, fixed point, C above O, upright rotation; small tilt perturbations.

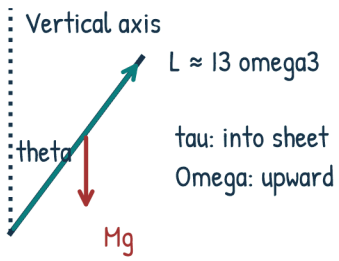
$$(I_3\omega_3)^2 > 4Mgd\,I_\perp$$

WHY / Linearised tilt equations give bounded oscillatory perturbations above this threshold; equality is marginal.

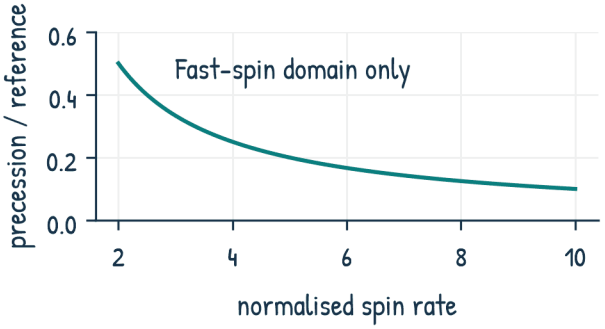
WATCH / No universal extra R/r factor exists. Real tip friction, damping and geometry require their own model.

Check E6: Omega≈0.25 rad/s; precession period≈25.13 s. Fast spin alone does not specify initial nutation.

Gravity turns spin angular momentum around vertical



Inverse-spin trend: do not extrapolate to zero spin



REMEMBER / Slow precession is a fast-spin approximation, not a small-tilt law; omega->0 is outside that approximation.

Flywheel limits: add a material law

Kinetic-energy storage is JEE core; elastic stress and failure models are specialised extensions.

Core energy, new restriction

$$K = \tfrac{1}{2} I\omega^2$$

$$\frac{K}{M} = \tfrac{1}{4}(\omega R)^2 \quad \text{(solid disc)}$$

For a prescribed spin rate this needs only rigid-body mechanics. A maximum allowable speed needs stress information.

WHY / Scaling to energy per mass removes size, but does not eliminate the need for a material strength limit.

WATCH / Do not call all flywheel energy questions "outside JEE"; it is the material theory that is additional.

Uniform elastic thin disc

Ideal isotropic constant-thickness disc, free outer edge, no centre hole, plane-stress linear elasticity.

$$\sigma_{\max} = \frac{3+\nu}{8}\rho\omega^2 R^2$$

$$\left(\frac{K}{M}\right)_{\max} = \frac{2\sigma_{\text{allow}}}{\rho(3+\nu)}$$

rho is density; nu is Poisson's ratio. Maximum stress is at the centre for ordinary positive nu.

WHY / Insert the stress-limited rim speed into K/M. Dimensions: stress/density = m²/s².

Numerical scale

rho=7800 kg/m³, nu=0.30, sigma(allow)=500 MPa; use the source's ideal strength as the allowable value.

$$(\omega R)_{\max} = 394.2 \text{ m/s}$$

$$K/M = 38.85 \text{ kJ/kg}$$

WHY / This independently recomputes the source's approximately 39 kJ/kg estimate.

WATCH / An engineering allowable stress usually includes safety factors, fatigue and manufacturing defects. This is not a safe design recommendation.

Changing shape changes stress

For an ideal thin rotating ring the hoop stress is rho omega² R², while K/M = omega² R² / 2.

$$(K/M)_{\max,\,ring} = \sigma_{\text{allow}}/(2\rho)$$

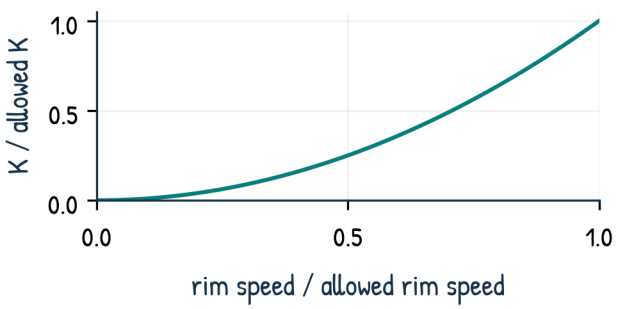
WHY / The ring stores more at fixed rim speed but reaches its own stress limit sooner; compare under the same failure condition.

WATCH / A bored disc, composite rotor, anisotropy, bearing mass or enclosure mass changes the estimate. A battery comparison is not a universal performance ranking.

Elastic disc model: no bore, isotropic, plane stress



Both kinetic energy and elastic stress scale as spin squared



REMEMBER / A perfectly rigid model cannot itself predict breaking speed: a constitutive and failure model must be supplied.

Open systems & missing constraints

Specialised model audit: not every printed question is sufficiently specified to have a unique answer.

Unrolling carpet: variable mass

A real roll has finite thickness t; a thin-layer continuum may approximate its cross-section. Negligible thickness literally cannot create a finite-radius roll.

$$m(R) \simeq MR^2/R_0^2$$

$$I_C(R) \simeq \tfrac{1}{2}m(R)R^2$$

WHY / With a negligible core and uniform packing, remaining mass scales with cross-sectional area.

WATCH / The rolled part is an open subsystem: momentum and energy flow out when layers are deposited.

What a complete model needs

Specify initial packing, core radius, thickness, release mechanism, no-slip contacts, and whether deposited material is stopped inelastically.

$$K_{\text{roll}} = \tfrac{3}{4}m(R)v_C^2$$

This expression assumes a uniform cylindrical roll and omega=v(C)/R. It is only an instantaneous kinetic energy, not an energy balance.

WHY / The whole-carpet energy must include changing potential, kinetic energy of every part, and any deposition loss.

WATCH / OL5 does not specify these laws. No unique speed-versus-radius result is asserted here.

Rod and peg: clarify the stages

A smooth floor supplies only vertical force; it cannot generally pin the rod's bottom. A peg contact is not automatically a permanent hinge.

$$\vec{L}_P^{\,-} \simeq \vec{L}_P^{\,+}$$

Across a short single-point peg impact this holds if other external angular impulses about fixed P are negligible.

WHY / Impulse at P has zero moment about P, but another simultaneous support impulse need not.

WATCH / OL9 lacks peg location/contact geometry, capture and restitution rules. Its assumed fixed-bottom fall contradicts a smooth floor.

After repairing the model

If a fixed pin captures the body at P, calculate pre-impact v(C) and omega, then L(P), then post-impact KE with the new inertia.

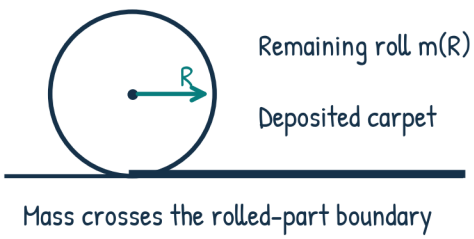
$$K_+ = \frac{L_P^2}{2I_P}$$

Only afterwards compare with the gravitational barrier. For unilateral contact, also test whether the peg reaction remains admissible.

WHY / This is the same phase logic used for the hinged bullet problem, but with new constraints explicitly supplied.

WATCH / No numerical swing-over condition can be certified from the source OL9 statement alone.

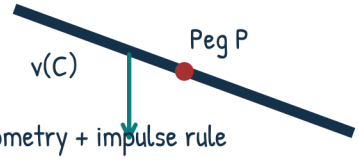
Open rolling subsystem: deposition needs a physical law



Mass crosses the rolled-part boundary

Peg impact is not automatically a new permanent hinge

Is contact captured or unilateral?



Geometry + impulse rule
+ later contact test

REMEMBER / If the model lacks a deposition law, contact geometry or impact rule, state that rather than inventing a final formula.